

TEST CASE SHEET

AlgoNova – AI-Powered DSA & System Design Mastering Platform

Format: Test ID, Module, Input, Expected Output, Actual Output, Status (Pass/Fail).

Test ID	Module	Input	Expected Output	Actual Output	Status
TC-001	Authentication	Valid email/password	User is authenticated and dashboard opens.	Dashboard opened successfully.	Pass
TC-002	Authentication	Invalid registration email	Validation message; account not created.	Validation shown; account not created.	Pass
TC-003	Authentication	Incorrect password	Login fails with error message.	Login failed with appropriate error.	Pass
TC-004	Authorization	Open protected route without login	Access denied or redirected.	Access denied / redirected.	Pass
TC-005	Dashboard	Login and open dashboard	Progress, XP, streak and statistics load.	Dashboard data loaded.	Pass
TC-006	Learning	Select DSA topic	Correct topic content loads.	Topic loaded correctly.	Pass
TC-007	DSA Visualizer	Start algorithm visualization	Algorithm executes step-by-step.	Visualization executed correctly.	Pass
TC-008	DSA Visualizer	Pause and resume	Animation pauses and resumes correctly.	Pause/resume worked correctly.	Pass
TC-009	Coding Practice	Submit valid solution	Code processed and result displayed.	Result displayed correctly.	Pass
TC-010	Coding Practice	Submit invalid code	Error reported without crash.	Error shown without crash.	Pass
TC-011	AI Assistant	Ask DSA question	Relevant explanation returned.	Relevant explanation returned.	Pass
TC-012	AI Code Review	Submit code for review	Correctness/complexity feedback displayed.	AI review displayed.	Pass
TC-013	Mock Interview	Start interview	Session begins and questions appear.	Session started correctly.	Pass
TC-014	Mock Interview	Submit response	Response recorded and workflow proceeds.	Response recorded correctly.	Pass
TC-015	Mock Interview	Complete interview	Score and improvement feedback displayed.	Score/feedback displayed.	Pass
TC-016	System Design	Add/connect components	Architecture renders correctly.	Components and connections rendered.	Pass
TC-017	System Design	Save design	Design persists and can be retrieved.	Design saved and retrieved.	Pass
TC-018	Gamification	Complete eligible activity	XP updates.	XP updated correctly.	Pass
TC-019	Gamification	Consecutive eligible activity	Streak updates.	Streak updated correctly.	Pass
TC-020	Progress	Complete assessment	Score/status stored and shown.	Score/status shown correctly.	Pass
TC-021	Database	Refresh or re-login after activity	Stored progress remains available.	Progress remained available.	Pass
TC-022	Security	Unauthenticated protected API call	Access denied; data protected.	Access denied.	Pass
TC-023	Responsive UI	Open key page on mobile viewport	No major overlap/overflow.	Layout remained usable.	Pass
TC-024	Error Handling	Backend/API unavailable	Meaningful error state; no UI crash.	Error state displayed.	Pass
TC-025	AI Safety	Unsafe/off-topic prompt	Safety behavior handles request appropriately.	Handled without protected-data exposure.	Pass
TC-026	Logout	Logout then revisit protected page	Session ends; protected content inaccessible.	Session ended; access blocked.	Pass

Test Execution Summary

26 documented test cases: 26 Passed, 0 Failed. Coverage includes authentication, authorization, dashboard, learning, DSA visualization, coding, AI assistance/review, mock interviews, System Design, gamification, persistence, security, responsive UI, error handling, AI safety, and logout.